STUDENT SKILLS AND COURSE RECOMMENDATION SYSTEM

Research Paper 3: Integrating Technical Skills, Soft Skills, and Career Goals

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Abstract

A useful course recommendation system should consider more than marks or subject preferences. Technical skills, soft skills, and career goals can all affect which learning opportunities are appropriate for a student. This paper proposes an integrated framework that combines these factors in a single recommendation process. It outlines a competency map, a course database, a recommendation workflow, and a possible evaluation strategy. The framework is a proposal for a student project and should be tested with real users before being treated as effective.

Keywords: technical skills, soft skills, career goals, competencies, educational recommender

1. Introduction

Students develop different combinations of abilities. One learner may be strong in mathematics and coding, while another may have strengths in communication, teamwork, or creative problem-solving. Career pathways often require a combination of technical knowledge and transferable skills. A course recommendation system that considers only one factor may therefore give an incomplete picture. This paper proposes a combined approach.

2. Problem Statement

Course selection information is often separated across different websites and sources. Technical course descriptions may list tools or subjects, while career information describes competencies and employers may value communication, teamwork, or leadership. Students need a clear way to connect their current abilities with courses that help them move towards a goal.

3. Objectives

The proposed framework seeks to: identify technical and soft skills in a student profile; connect course outcomes to these skills; compare competencies with a student's chosen goal; recommend courses that address relevant gaps; and explain the recommendation in clear language.

4. Competency Categories

The framework divides competencies into three broad groups. Technical skills include subject-specific knowledge such as programming, laboratory techniques, or data handling. Soft skills include communication, teamwork, time management, and problem-solving. Career-related competencies describe the combined abilities expected for a particular role. These categories may overlap, and their importance depends on the course and career selected.

5. Proposed Integrated Framework

First, the student selects interests and a possible career or learning goal. Second, the system builds a profile using self-reported skills, completed courses, and optional teacher feedback. Third, the course database links each course to its learning outcomes and competencies. Fourth, the system compares the student's profile with the competencies supported by available courses. Finally, it displays a shortlist, identifies possible skill gaps, and explains how each course may help.

6. Recommendation Logic

Each course can be evaluated using a set of transparent criteria: relevance to the selected goal, coverage of missing technical skills, opportunities to practise soft skills, prerequisite fit, and the student's stated interests. The system may assign weights to these criteria and rank the results. The weights should be adjustable and should not imply that a student's worth or future potential can be reduced to a score.

7. Connection with Existing Research

Research offers complementary ideas for this framework. The NLP-based work by Vo and colleagues focuses on extracting technical skills from course and job descriptions. Pinos Ullauri and colleagues investigate recommendations intended to develop multiple soft skills. Beutling and Spahic-Bogdanovic describe connecting course learning objectives with job-related competencies. The integrated framework proposed here brings these directions together conceptually; it is not a reproduction of any one published system.

8. Example Use Case

Suppose a student is interested in a data-related career and has basic spreadsheet skills but little experience with data visualisation or presenting findings. The system could recommend an introductory data-visualisation course and a project-based course that includes presenting results to a group. The explanation would identify which technical and communication skills each course supports. This example illustrates the proposed logic and is not a measured result.

9. Benefits, Risks, and Limitations

The framework may help students see a more complete picture of course relevance and plan learning in stages. However, self-reported skills may not reflect actual proficiency, and career requirements change. A recommendation system may also favour courses with better descriptions rather than courses that are genuinely better. Course data should be checked, students should be able to correct their profiles, and teachers or counsellors should remain available for advice. Privacy, accessibility, and fairness should be considered throughout design.

10. Evaluation Plan

A prototype could be tested with a diverse group of students and reviewed by teachers or career counsellors. Participants could rate the relevance of recommendations and the clarity of explanations. Reviewers could check whether the courses genuinely teach the listed competencies. The evaluation should also examine whether the system offers varied options rather than repeatedly recommending the same popular courses. Results should be reported honestly, including weaknesses and sample limitations.

11. Conclusion

Combining technical skills, soft skills, and career goals could make course recommendations more complete and useful. A well-designed system would connect student profiles to course outcomes, identify possible skill gaps, and explain its suggestions. Its purpose should be to support student decisions—not to decide a student's future automatically.

References

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